

Copy of AP Calc Unit 1 Test

Name _____

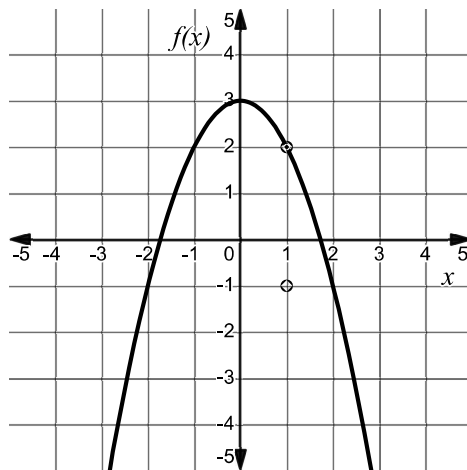
1. The graph of f is shown. Evaluate $\lim_{x \rightarrow 1} f(x)$.

A) The limit does not exist.

B) -1

C) 2

D) 3



2. The function f has the following behavior: $\lim_{x \rightarrow \infty} f(x) = 4$ and $\lim_{x \rightarrow -\infty} f(x) = 2$.

Which of the following statements is not guaranteed to be true?

A) The function f has a horizontal asymptote at $y = 4$.

B) The function f has a horizontal asymptote at $y = 2$.

C) The function f has different end behaviors as x approaches positive and negative infinity.

D) The function f has two vertical asymptotes at $x = 2$ and $x = 4$.

3. For which of the following functions is $\lim_{x \rightarrow 2} g(x) = 0$?

A) $g(x) = \frac{x-2}{x-3}$

B) $g(x) = \frac{x+2}{x-3}$

C) $g(x) = \frac{\sin(x)}{x}$

D) $g(x) = \frac{|2x-2|}{x-1}$

4. If f is a function with infinite discontinuities at $x = -4$ and $x = 1$, which of the following could be the equation of f ?

A) $f(x) = \frac{x^2 + 3x - 4}{(x^2 - 1)(x + 4)}$

B) $f(x) = \frac{x^2 + 3x - 4}{(x^2 - 4)(x - 1)}$

C) $f(x) = \frac{x^2 - 1}{x^2 + 3x - 4}$

D) $f(x) = \frac{x^2 - 4}{x^2 + 3x - 4}$

5. Let $f(x) = \begin{cases} x^2 - a^2x & x < 2 \\ 4 - 2x^2 & x \geq 2 \end{cases}$

For what value(s) of a , if any, is f continuous at $x = 2$?

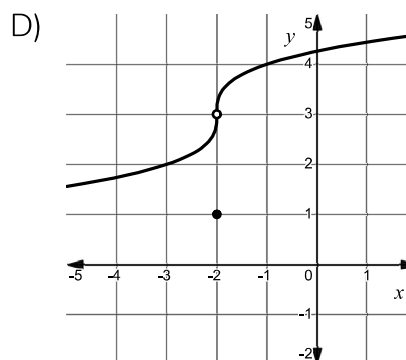
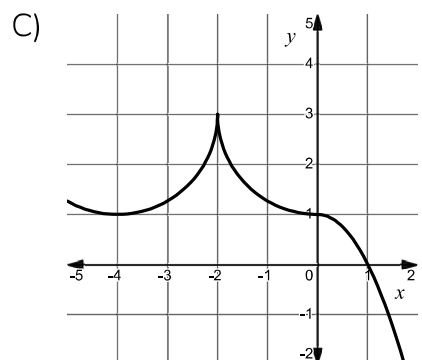
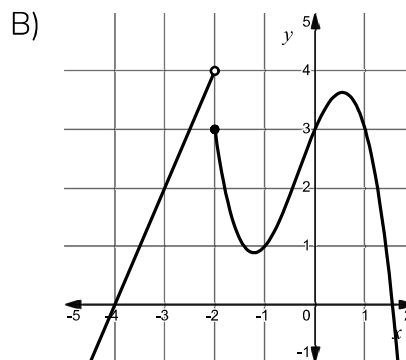
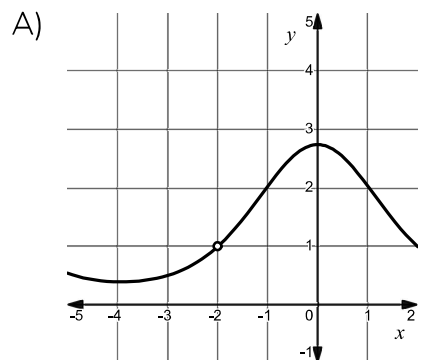
A) 2 only

B) 2 or -2

C) -2 only

D) No such value exists

6. Which of the following could be the graph of a function h such that $\lim_{x \rightarrow -2} h(x)$ does not exist?



7. Let $\lim_{x \rightarrow 1} f(x) = -2$, $\lim_{x \rightarrow 2} f(x) = -5$, $\lim_{x \rightarrow 1} g(x) = -8$, and $\lim_{x \rightarrow 2} g(x) = 16$ be true for functions f and g .

Evaluate $\lim_{x \rightarrow 1} \left[\frac{3f(2x)}{2g(x)} \right]$.

A) $-\frac{3}{8}$

B) $\frac{3}{8}$

C) $\frac{15}{16}$

D) $-\frac{15}{16}$

8. Let functions f and g be given by the equations $f(x) = 2x^2 + 5x - 12$ and $g(x) = 20 + 9x + x^2$.

Evaluate $\lim_{x \rightarrow -4} \frac{f(x)}{g(x)}$.

A) 11

B) $-\frac{1}{11}$

C) -11

D) Does not exist

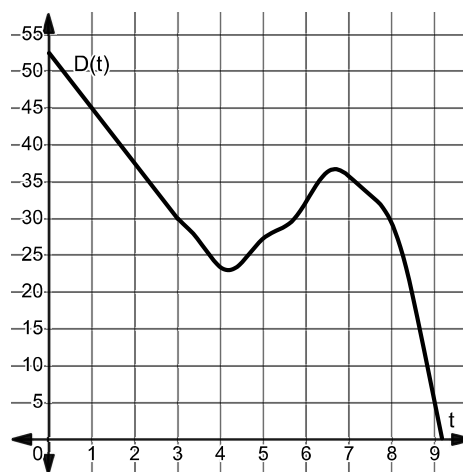
9. The height of a drone can be modeled by the function D , where $D(t)$ is measured in feet and t is the time in minutes. The graph of D is given. What is the best approximation of the rate at which height of the drone is changing at $t = 2$ minutes?

A) $-2 \frac{\text{ft}}{\text{min}}$

B) $-7.5 \frac{\text{ft}}{\text{min}}$

C) $-15 \frac{\text{ft}}{\text{min}}$

D) $-37.5 \frac{\text{ft}}{\text{min}}$



10. It is known that for continuous functions g and h , $g(x) \leq f(x) \leq h(x)$ for $-2 < x < 2$. It is also known that $\lim_{x \rightarrow 0} g(x) = \lim_{x \rightarrow 0} h(x) = -1$. Which of the following statements is NOT guaranteed?
- A) $\lim_{x \rightarrow 1} f(x) = \lim_{x \rightarrow 1} g(x)$
- B) $\lim_{x \rightarrow 0} f(x) = -1$
- C) $\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^+} f(x)$
- D) $g(0) = -1$
11. Let f be the function given by $f(x) = \sin\left(\frac{\pi}{2}x\right) + 3x$. The Intermediate Value Theorem applied to f on the closed interval $[1, 3]$ guarantees a solution on $[1, 3]$ to which of the following equations?
- A) $f(x) = 0$
- B) $f(x) = 2$
- C) $f(x) = 6$
- D) $f(x) = 9$
12. $\lim_{x \rightarrow \infty} \frac{12 + 5x^3}{3 - 2e^x} =$
- A) $-\frac{5}{2}$
- B) 0
- C) 4
- D) The limit does not exist.

13. The piecewise function f is defined below. The graph of a piecewise function $y = h(x)$ and selected values of a continuous function g are given.

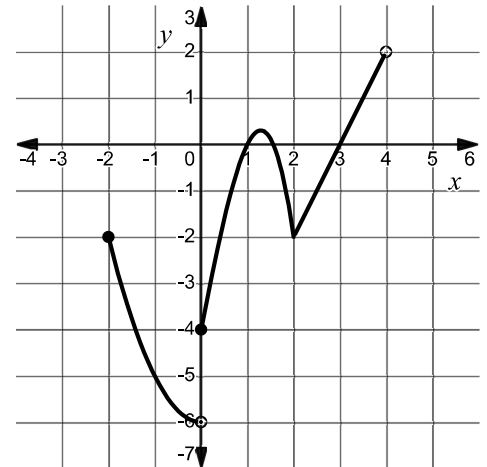
$$\text{Let } f(x) = \begin{cases} 3x + 4 & x < -2 \\ h(x) & -2 \leq x < 4 \\ g(x) & 4 \leq x \leq 8 \\ \frac{-3x + cx^2}{2(x - 10)(x + 4)} & x > 8 \end{cases}$$

x	4	5	6	7	8
$g(x)$	3	1	-1	-3	-5

- a. At which x -value(s) on the interval $(-2, 4)$ does the graph of h have a removable discontinuity?

- b. If $\lim_{x \rightarrow \infty} f(x) = 6$, find the value of c .

- c. Is f continuous at $x = -2$? Justify your answer.



- d. Assuming the value of c found in part b, at which x -values does the graph of f have an infinite discontinuity? Give reasoning to support your answer.

- e. What is the minimum number of solutions to $f(x) = 0$ for $-2 \leq x \leq 8$? Justify your answer.

14. The function f is continuous for all values of x except $x = 2$. It is known that $f(2) = 6$. Values of $f(x)$ at selected values of x are given in the table below.

x	1.9	1.99	1.999	2.001	2.01	2.1	4.9	4.99	4.999	5.001	5.01	5.1
$f(x)$	2.52	2.79	2.982	3.018	3.21	3.48	8.65	8.342	8.122	7.878	7.658	7.35

- a. Write, but do not evaluate, an expression that gives an approximation for the instantaneous rate of change of f at $x = 5$.
- b. What type of discontinuity does f have at $x = 2$? Justify your answer.
- c. Use the table to approximate $\lim_{x \rightarrow 2} 3f(x) - \lim_{x \rightarrow 5} \frac{f(x)}{2}$.